RETAIL SALES & BUSINESS PERFORMANCE ANALYSIS

Business Findings & Recommendations Report | Sample Superstore

Purpose: Summarize patterns observed in the supplied retail dataset and suggest practical actions. Findings describe the data; they do not establish causes.

1. Executive Summary

Technology was the strongest category by total sales and profit. The West region was the strongest region by total sales and profit. Within Furniture, Tables had negative average profit while Chairs had positive average profit. These patterns suggest opportunities to review product profitability, compare regional performance, and investigate monthly sales patterns.

2. Data Overview

Item	Details
Dataset	Sample Superstore
Cleaned records	10,195 rows
Date range	2023–2026 (confirmed for this project)
Key fields	Order Date, Sales, Profit, Quantity, Category, Sub-Category, Region, Customer Segment, Product Name, Order ID

3. Methodology

The dataset was inspected and cleaned using Python and pandas. Cleaning included checking and handling missing values, removing duplicate rows, and verifying/correcting data types. Exploratory analysis and charts were produced with pandas and Matplotlib. Business summaries were queried in MySQL. An interactive Power BI dashboard was built with KPI cards, trend and comparison visuals, and Order Date, Category, and Region slicers. Month-Year and Month-Year Sort calculated columns were added in Power BI for monthly display and chronological sorting. KPI cards use default Sum and Distinct Count aggregations rather than explicit DAX measures.

4. Business Findings

1. Category performance

Technology generated the highest total sales and total profit among the three categories.

2. Regional performance

The West region recorded the highest total sales and total profit among the four regions.

3. Furniture sub-category profitability

Tables had the lowest average profit (-54.46), while Chairs had the highest (+42.94). Bookcases averaged -15.66. Tables also had the highest average discount (25.81%); this does not prove discounts caused lower profit.

4. Customer segment performance

Consumer customers generated approximately 1.17M in sales and 136,371.45 in profit. Earlier analysis notes list Corporate second and Home Office lowest; recheck this ranking against the source summary before publication.

5. Monthly sales pattern

September–December frequently showed stronger monthly sales, while January–February frequently showed lower sales. November 2026 was highest (118,454.51), and February 2023 was lowest (4,519.89). This does not prove seasonality or its cause.

6. Top product by sales

Canon imageCLASS 2200 Advanced Copier ranked highest by total sales among the ten products in the top-10 sales analysis.

5. Business Recommendations

1. Review Furniture Tables profitability. Because Tables had negative average profit (-54.46), review pricing, costs, discount levels, and product mix. Validate changes by comparing product-level profit and margin before and after a controlled pricing or discount test.

2. Investigate regional performance differences. Because West led total sales and profit, compare its product mix, order volume, customer segments, and discount patterns with other regions. Test whether practices associated with West can be adapted elsewhere, then monitor regional sales and profit.

3. Plan for observed monthly demand patterns. Because September–December frequently showed stronger sales, review inventory availability and marketing plans ahead of those months. Compare future monthly sales and stock availability to see whether the pattern repeats.

6. Limitations & Assumptions

• The analysis describes historical records and does not predict or guarantee future performance.

• Comparisons do not prove causation. The dataset does not establish that discounts caused changes in sales or profit.

• The September–December pattern may indicate seasonality, but additional years and external context are needed to validate it.

• The available fields do not provide a complete view of operating expenses or all cost drivers.

• Customer segment comparisons describe recorded sales and profit; they do not explain customer motivations.

• KPI cards use implicit Power BI aggregations. Order count is the distinct count of Order ID; dashboard values are rounded.

7. KPI Validation

Metric	MySQL	Power BI	Check
Total Sales	2,326,534.35	2.33M	Matched
Total Profit	292,296.81	292.30K	Matched
Total Orders	5,111	5.111K	Matched
Total Quantity	38,654	39K	Matched

8. Appendix — Reproducibility

Supporting materials: notebooks/retail_sales_analysis.ipynb; SQL analysis files in sql/; data/processed/superstore_cleaned.csv; and dashboard/Retail_Sales_Dashboard.pbix. Read this report alongside those materials.

Publication check: Recheck the Home Office segment ranking and confirm the data source's sharing permissions before sharing. Do not publish restricted raw data or credentials.